

8085 TIMING DIAGRAMS OF INSTRUCTIONS

Special Note:

Any operation performed INSIDE the microprocessor does not require a machine cycle and hence will not be shown in timing diagrams.

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1) MVI B, 25H

$B \leftarrow 25H$

Machine Cycle	Address Bus	Data Bus	T-States
Opcode Fetch	PC	Opcode (Eg: 3E)	4
Memory Read	PC + 1	25	3
Total			7

2) LXI B, 2000H

$BC \leftarrow 2000H$

Machine Cycle	Address Bus	Data Bus	T-States
Opcode Fetch	PC	Opcode (Eg: 3E)	4
Memory Read	PC + 1	00	3
Memory Read	PC + 2	20	3
Total			10

3) LDA 2000H

$A \leftarrow [2000H]$

Machine Cycle	Address Bus	Data Bus	T-States
Opcode Fetch	PC	Opcode (Eg: 3E)	4
Memory Read	PC + 1	00 (Z)	3
Memory Read	PC + 2	20 (W)	3
Memory Read	2000	[2000]	3
Total			13



4) STA 3000H

$A \rightarrow [3000H]$

Machine Cycle	Address Bus	Data Bus	T-States
Opcode Fetch	PC	Opcode (Eg: 3E)	4
Memory Read	PC + 1	00 (Z)	3
Memory Read	PC + 2	30 (W)	3
Memory Write	3000	A	3
Total			13

5) LDAX B

$A \leftarrow [BC]$

Machine Cycle	Address Bus	Data Bus	T-States
Opcode Fetch	PC	Opcode (Eg: 3E)	4
Memory Read	BC	[BC]	3
Total			7

6) STAX D

$A \rightarrow [DE]$

Machine Cycle	Address Bus	Data Bus	T-States
Opcode Fetch	PC	Opcode (Eg: 3E)	4
Memory Write	DE	A	3
Total			7

7) LHLD 2000H

$L \leftarrow [2000H], H \leftarrow [2001H]$

Machine Cycle	Address Bus	Data Bus	T-States
Opcode Fetch	PC	Opcode (Eg: 3E)	4
Memory Read	PC + 1	00 (Z)	3
Memory Read	PC + 2	20 (W)	3
Memory Read	2000	[2000]	3
Memory Read	2001	[2001]	3
Total			16

8) SHLD 5140H

L → [5140H], H → [5141H]

Machine Cycle	Address Bus	Data Bus	T-States
Opcode Fetch	PC	Opcode (Eg: 3E)	4
Memory Read	PC + 1	40 (Z)	3
Memory Read	PC + 2	51 (W)	3
Memory Write	5140	L	3
Memory Write	5141	H	3
Total			16

9) MOV B,C

B ← C

Machine Cycle	Address Bus	Data Bus	T-States
Opcode Fetch	PC	Opcode (Eg: 3E)	4
Total			4

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10) PCHL

$PC \leftarrow HL$

6T

Opcode Fetch

Machine Cycle	Address Bus	Data Bus	T-States
Opcode Fetch	PC	Opcode (Eg: 3E)	6
Total			6

11) SPHL

$SP \leftarrow HL$

6T

Opcode Fetch

Machine Cycle	Address Bus	Data Bus	T-States
Opcode Fetch	PC	Opcode (Eg: 3E)	6
Total			6

12) ADD B {all 8 bit arithmetic operations using register addressing mode}

$A \leftarrow A + B$

Machine Cycle	Address Bus	Data Bus	T-States
Opcode Fetch	PC	Opcode (Eg: 3E)	4
Total			4



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13) INR B

$B \leftarrow B + 1$

Machine Cycle	Address Bus	Data Bus	T-States
Opcode Fetch	PC	Opcode (Eg: 3E)	4
Total			4

14) INX B

$BC \leftarrow BC + 1$

6T

Opcode Fetch

Machine Cycle	Address Bus	Data Bus	T-States
Opcode Fetch	PC	Opcode (Eg: 3E)	6
Total			6



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INSTRUCTIONS INVOLVING M – MEMORY POINTER

Special Note:

- Instructions involving M must be examined more carefully.
- Remember M is not a register. M is a MEMORY LOCATION pointed by HL pair.
- Taking data from M will need a Memory Read cycle.
- Putting data in M will need a Memory Write cycle.

15) MVI B, 25H

$B \leftarrow B + 1$

Machine Cycle	Address Bus	Data Bus	T-States
Opcode Fetch	PC	Opcode (Eg: 3E)	4
Memory Read	PC + 1	25	3
Memory Write	HL	25	3
Total			10

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16) MOV B, M

$B \leftarrow M$

Machine Cycle	Address Bus	Data Bus	T-States
Opcode Fetch	PC	Opcode (Eg: 3E)	4
Memory Read	HL	M	3
Total			7

17) MOV M, B

$M \leftarrow B$

Machine Cycle	Address Bus	Data Bus	T-States
Opcode Fetch	PC	Opcode (Eg: 3E)	4
Memory Write	HL	B	3
Total			7

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18) INR M {Very Important}

$$M \leftarrow M + 1$$

Machine Cycle	Address Bus	Data Bus	T-States
Opcode Fetch	PC	Opcode (Eg: 3E)	4
Memory Read	HL	M	3
Memory Write	HL	M + 1	3
Total			10

19) ADD M

$$A \leftarrow A + M$$

Machine Cycle	Address Bus	Data Bus	T-States
Opcode Fetch	PC	Opcode (Eg: 3E)	4
Memory Read	HL	M	3
Total			7

STACK OPERATIONS

20) PUSH B

$SP \leftarrow SP - 1$... internal operation
 $[SP] \leftarrow B$... memory write
 $SP \leftarrow SP - 1$... internal operation
 $[SP] \leftarrow C$... memory write

6T
Opcode Fetch

Machine Cycle	Address Bus	Data Bus	T-States
Opcode Fetch	PC	Opcode (Eg: 3E)	6
Memory Write	SP - 1	B	3
Memory Write	SP - 2	C	3
Total			12

21) POP B

$C \leftarrow [SP]$... memory read
 $SP \leftarrow SP + 1$... internal operation
 $B \leftarrow [SP]$... memory read
 $SP \leftarrow SP + 1$... internal operation

Machine Cycle	Address Bus	Data Bus	T-States
Opcode Fetch	PC	Opcode (Eg: 3E)	4
Memory Read	SP	[SP]	3
Memory Read	SP + 1	[SP + 1]	3
Total			10

BRANCH INSTRUCTIONS

22) JMP 2000H

PC $\leftarrow$ 2000 H

Machine Cycle	Address Bus	Data Bus	T-States
Opcode Fetch	PC	Opcode (Eg: 3E)	4
Memory Read	PC + 1	00 (Z)	3
Memory Read	PC + 2	20 (W)	3
Total			10

23) JC 2000H

If CF = 1 then condition is true hence,

Machine Cycle	Address Bus	Data Bus	T-States
Opcode Fetch	PC	Opcode (Eg: 3E)	4
Memory Read	PC + 1	00 (Z)	3
Memory Read	PC + 2	20 (W)	3
Total			10

If CF = 0 then condition is false hence,

Machine Cycle	Address Bus	Data Bus	T-States
Opcode Fetch	PC	Opcode (Eg: 3E)	4
Memory Read (Idle)	---	---	3
Total			7



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24) Call 2000H

$SP \leftarrow SP - 1$... internal operation
 $[SP] \leftarrow PCH$... Memory Write
 $SP \leftarrow SP - 1$... internal operation
 $[SP] \leftarrow PCL$... Memory Write
 $PC \leftarrow 2000 H$... internal operation

6T
Opcode Fetch

Machine Cycle	Address Bus	Data Bus	T-States
Opcode Fetch	PC	Opcode (Eg: 3E)	6
Memory Read	PC + 1	00 (Z)	3
Memory Read	PC + 2	20 (W)	3
Memory Write	SP - 1	PCH	3
Memory Write	SP - 2	PCL	3
Total			18

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25) CC 2000H

If CF=1 then condition is true hence,

$SP \leftarrow SP - 1$... internal operation
 $[SP] \leftarrow PCH$... Memory Write
 $SP \leftarrow SP - 1$... internal operation
 $[SP] \leftarrow PCL$... Memory Write
 $PC \leftarrow 2000H$... internal operation

6T
Opcode Fetch

Machine Cycle	Address Bus	Data Bus	T-States
Opcode Fetch	PC	Opcode (Eg: 3E)	6
Memory Read	PC + 1	00 (Z)	3
Memory Read	PC + 2	20 (W)	3
Memory Write	SP - 1	PCH	3
Memory Write	SP - 2	PCL	3
Total			18

If CF = 0 then condition is false hence,

Machine Cycle	Address Bus	Data Bus	T-States
Opcode Fetch	PC	Opcode (Eg: 3E)	6
Memory Read (Idle)	---	---	3
Total			9

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26) RET

PCL $\leftarrow$ [SP] ... Memory Read
SP $\leftarrow$ SP + 1 ... internal operation
PCH $\leftarrow$ [SP] ... Memory Read
SP $\leftarrow$ SP + 1 ... internal operation

Machine Cycle	Address Bus	Data Bus	T-States
Opcode Fetch	PC	Opcode (Eg: 3E)	4
Memory Read	SP	[SP]	3
Memory Read	SP + 1	[SP + 1]	3
Total			12



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27) RC

If $CF = 1$ then condition is true hence,

$PCL \leftarrow [SP]$... Memory Read
 $SP \leftarrow SP + 1$... internal operation
 $PCH \leftarrow [SP]$... Memory Read
 $SP \leftarrow SP + 1$... internal operation

6T

Opcode Fetch

Machine Cycle	Address Bus	Data Bus	T-States
Opcode Fetch	PC	Opcode (Eg: 3E)	6
Memory Read	SP	[SP]	3
Memory Read	SP + 1	[SP + 1]	3
Total			12

If $CF = 0$ then condition is false hence,

Machine Cycle	Address Bus	Data Bus	T-States
Opcode Fetch	PC	Opcode (Eg: 3E)	6
Total			9

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28) RSTn

SP $\leftarrow$ SP - 1 ... internal operation
[SP] $\leftarrow$ PCH ... Memory Write
SP $\leftarrow$ SP - 1 ... internal operation
[SP] $\leftarrow$ PCL ... Memory Write
PC $\leftarrow$ (n x 8) ... internal operation

6T
Opcode Fetch

Machine Cycle	Address Bus	Data Bus	T-States
Opcode Fetch	PC	Opcode (Eg: 3E)	6
Memory Write	SP - 1	PCH	3
Memory Write	SP - 2	PCL	3
Total			12



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I/O OPERATIONS

29) IN 80H

$A \leftarrow [80]_{I/O}$

Machine Cycle	Address Bus	Data Bus	T-States
Opcode Fetch	PC	Opcode (Eg: 3E)	4
Memory Read	PC + 1	80	3
I/O Read	80	[80]	3
Total			10

30) OUT 80H

$A \rightarrow [80]_{I/O}$

Machine Cycle	Address Bus	Data Bus	T-States
Opcode Fetch	PC	Opcode (Eg: 3E)	4
Memory Read	PC + 1	80	3
I/O Write	80	A	3
Total			10

ADDITIONAL INSTRUCTIONS

31) DAD D

$HL \leftarrow HL + DE$

Machine Cycle	Address Bus	Data Bus	T-States
Opcode Fetch	PC	Opcode (Eg: 3E)	4
Bus Idle	---	---	3
Bus Idle	---	---	3
Total			10

32) HLT

$\text{Halt F/F} \leftarrow 1$

Machine Cycle	Address Bus	Data Bus	T-States
Opcode Fetch	PC	Opcode (Eg: 3E)	4T + 1T
Total			5

33) XTHL

$Z \leftarrow [SP]$... Memory Read
 $W \leftarrow [SP + 1]$... Memory Read
 $[SP + 1] \leftarrow H$... Memory Write
 $[SP] \leftarrow L$... Memory Write
 $HL \leftarrow WZ$... internal operation

Machine Cycle	Address Bus	Data Bus	T-States
Opcode Fetch	PC	Opcode (Eg: 3E)	4
Memory Read	SP	[SP]	3
Memory Read	SP + 1	[SP + 1]	3
Memory Write	SP + 1	H	3
Memory Write	SP	L	3
Total			16

34) XCHG

$DE \leftrightarrow HL$... internal operation

Machine Cycle	Address Bus	Data Bus	T-States
Opcode Fetch	PC	Opcode (Eg: 3E)	4
Total			4



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Timing Diagram for Call 2000 H

